



## Introduction

Financial markets are highly sensitive to political information. Donald J. Trump, the President of the United States, is a highly influential political figure known for his intensive use of Truth Social to express political views and policy positions. Given his substantial influence on public expectations, we hypothesize that Trump’s social media posts contain emotional and informational signals that may affect investors’ beliefs and thus predict short-term stock market movements. Motivated by this idea, this study builds a simple text-based prediction model that uses Trump’s daily Truth Social posts to forecast whether the S&P 500 will go up or down on the following trading day. Using natural language processing and supervised machine learning methods, we evaluate three Naive Bayes classifiers (Multinomial, Gaussian, and Bernoulli) as well as Logistic Regression. Our empirical results show that Bernoulli Naive Bayes (BNB) performs best among all models, achieving a prediction accuracy of 72% and a higher F1-score (0.62) in predicting market upturns compared with alternative specifications. In contrast, the predictive power of Logistic Regression is relatively modest in this small-sample, high-dimensional setting.

## Proposed Model

### Naive Bayes

The Naive Bayes classifier estimates the posterior probability of a market movement label  $y \in \{0, 1\}$  given the TF-IDF feature vector  $\mathbf{x}_d$  of Trump’s daily posts, using Bayes’ rule

$$P(y | \mathbf{x}_d) \propto P(y) \prod_{i=1}^V P(x_{i,d} | y)$$

$$1. \text{Multinomial Naive Bayes } P(t | y) = \frac{N_{t,y} + \alpha}{\sum_k N_{k,y} + \alpha V}$$

$$2. \text{Gaussian Naive Bayes } P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_{i,y}^2}} \exp\left(-\frac{(x_i - \mu_{i,y})^2}{2\sigma_{i,y}^2}\right)$$

$$3. \text{Bernoulli Naive Bayes } P(x_i = 1 | y) = \frac{N_{i,y} + \alpha}{N_y + 2\alpha}$$

### Logistic Regression

Logistic Regression is employed as an alternative classification method. Its conditional probability is modeled as:

$$P(y_d = 1 | \mathbf{x}_d) = \frac{1}{1 + e^{-(\beta_0 + \mathbf{x}_d^T \beta)}}$$

The predicted class is obtained by:

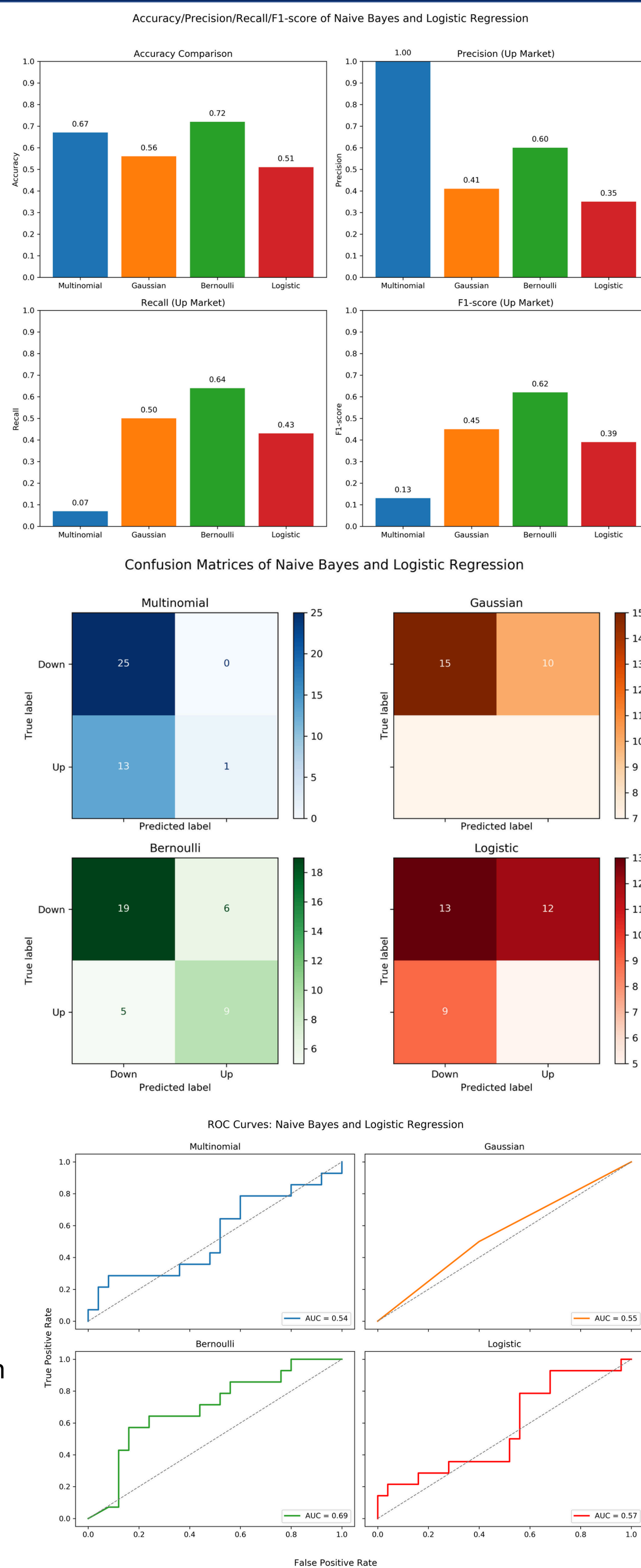
$$\hat{y}_d = \begin{cases} 1, & \text{if } P(y_d = 1 | \mathbf{x}_d) \geq 0.5 \\ 0, & \text{otherwise} \end{cases}$$

### Random Splitting vs Time-Based Splitting

Time-series split (first 80% days for training, last 20% days for testing) to prevent look-ahead bias (Glasserman & Lin, 2024).

### Model Selection

- Accuracy:** Overall correctness of the model.
- Precision/Recall/F1-Score:** Evaluate classification performance under class imbalance, with a special emphasis on its ability to capture profitable “Up” signals.
- Confusion matrix:** Shows the structure of prediction errors.
- AUC (Area Under the ROC Curve):** Ability of the model to distinguish between “Up” and “Down”.



## Experimental Setup

### Datasets

- Social Media:** ~4,700 posts from Trump on Truth Social.
- Financial:** Daily S&P 500 stock movement (labeled as 1 for Up, 0 for Down).
- Time Span:** January 22 to October 24, 2025
- Dataset Construction:** Merge daily social media data with financial data, handling non-trading days.

### Data Preprocessing

- Text Cleaning:** Removal of URLs, HTML tags, and non-alphabetic characters.
- Vectorization:** Converted cleaned text into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency) to capture the importance of specific political keywords.

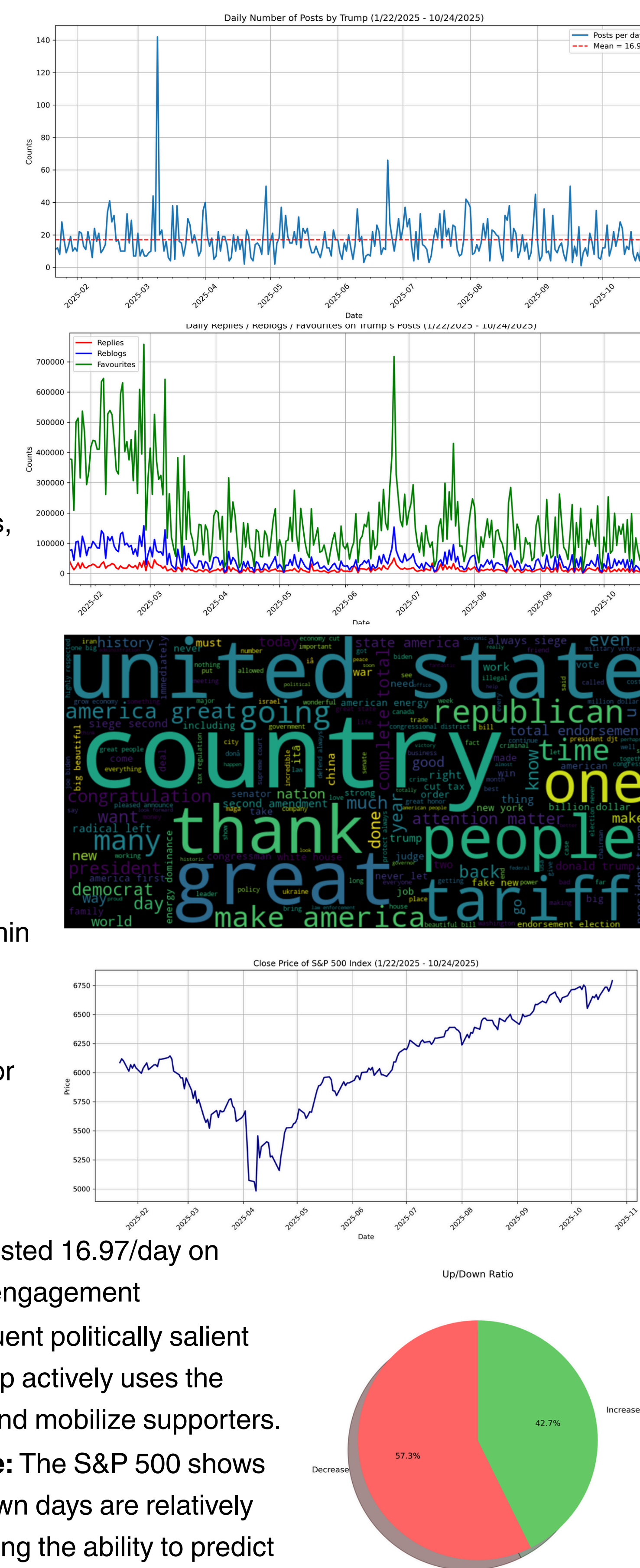
$$TF-IDF_{i,d} = TF_{i,d} \times IDF_i$$

$$TF_{i,d} = \frac{n_{i,d}}{\sum_k n_{k,d}}$$
 captures the relative importance of a word within a single day's political communication

$$IDF_i = \log\left(\frac{N}{df_i}\right)$$
 reflects how informative or rare a word is across the entire sample period

### Visualization

- High posting activity & influence:** Trump posted 16.97/day on average, with consistently high levels of user engagement
- Text features:** The word cloud highlights frequent politically salient words such as “republican,” showing that Trump actively uses the platform to communicate political messages and mobilize supporters.
- Market variation with mild class imbalance:** The S&P 500 shows substantial daily fluctuations. While up and down days are relatively balanced, down days account for 57.3%, making the ability to predict market increases a key criterion for model evaluation.



## Results and Conclusions

### Results

- Bernoulli Naive Bayes (BNB) outperformed all other models with an Accuracy of 72% and an F1-score (up) of 0.62.
- Multinomial Naive Bayes had high precision but very low recall (it rarely predicted "Up").
- Logistic Regression performed poorly, barely better than a coin flip.

#### Extract Top 10 Up/Down words

$$\text{Log-Odds}(t) = \log P(t | y = 1) - \log P(t | y = 0)$$

If  $\text{Log-Odds}(t) > 0$ , the word is more strongly associated with market increases ("up").

If  $\text{Log-Odds}(t) < 0$ , the word is more strongly associated with market decreases ("down").

| Top 10 UP Words  | Top 10 DOWN Words |
|------------------|-------------------|
| state missouri   | answer            |
| controlled       | get vote          |
| sadly            | great big         |
| many thing       | wealth            |
| missouri         | dishonest         |
| we agreed        | finish            |
| backing          | stock market      |
| order help       | prayer            |
| mr               | patronis          |
| support military | we making         |

### Discussion & Conclusion

#### 1. Hypothesis Validation:

This study empirically validates that Donald Trump's social media posts on Truth Social contain latent emotional and informational signals that effectively predict short-term movements of the S&P 500 index.

#### 2. Model Performance:

Bernoulli Naive Bayes (BNB) emerged as the superior model, achieving an Accuracy of 72% and an AUC of 0.69. The confusion matrices reveal that BNB provides the most balanced classification, whereas other models struggled with high false-negative rates (failing to predict market upturns).

#### 3. Interpretability & Insights:

The success of the Bernoulli model (which uses binary feature vectors) suggests that the presence or absence of specific political keywords is a stronger predictor of market reaction than the frequency of those words.

#### 4. Limitations & Future Directions:

- The sample is relatively small (192 trading days); future studies should use longer time spans.
- Only daily market data are used; future work can explore minute-level or intraday reactions.
- Current features rely on TF-IDF; future research may apply sentiment analysis or deep learning NLP models.

### Reference

Glasserman, P., & Lin, S. (2024). Assessing look-ahead bias in stock return predictions generated by GPT sentiment analysis. Journal of Financial Data Science, 6(1), 25–42.